Xuanzhi Chen

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Website | Google Scholar | GitHub | LinkedIn

Research Interests

Spatial multi-omics, machine learning for biology, and scientific agents.

Education

Peking University Beijing, China

B.Eng. Candidate in Bioinformatics, School of Life Sciences *Sep. 2023 – Jun. 2027 (Expected)*

- **GPA: 3.77/4.0.** Pursuing a double degree in Economics at the National School of Development.
- **Selected coursework:** Computational Immuno-oncology (99/100); Data Structures and Algorithms (98/100); Statistical Analysis of Genomics Data (A); AI Methods and Applications in Life Sciences (A); Fundamentals of Network Science and Systems Biology (A).

Preprint

Nianping Liu, **Xuanzhi Chen**, Miao Cui, Xiaoying Liao, Xiaoke Song, Sairam Pantham, Weize Xu, and Xiaojie Qiu. **Bio-Babel: autonomous cross-language reconstruction of agent-ready computational biology ecosystems.** *bioRxiv*, 2026. doi:10.64898/2026.08.30.743167.

Research Experience

College of Future Technology, Peking University Beijing, China

Undergraduate Researcher; Advisor: Prof. Hu Zeng *Jul. 2024 – Present*

- **STATES-GBM:** Built and optimized the upstream spatial omics pipeline for glioblastoma, focusing on cell segmentation and image stitching; performed malignant-cell identification and subtyping and spatial analysis of translation efficiency.
- **iST Agent:** Developed a contract- and memory-guided agent for imaging-based spatial transcriptomics. Generated machine-readable contracts for seven pipelines and reproduced a laboratory pipeline's decoded output byte for byte.

Department of Genetics, Stanford University Stanford, CA

Undergraduate Researcher; Advisor: Prof. Xiaojie Qiu *Jun. 2026 – Present*

On-site summer research, Jun.–Aug. 2026; continued remotely.

- **Bio-Babel:** Built an annotation agent to generate machine-readable usage contracts for reconstructed packages; extended coding-agent compatibility and Python-to-C++ reconstruction. Conducted Bio-Babel's pancreatic differentiation case study, reviewing agent-executed analyses and completing genotype-aware screening and biological interpretation of reduced maturation signatures under SWI/SNF loss.
  Code: Pancreas case study | UMI-tools-cpp.
- **End-to-end spatial transcriptomics modeling:** Developing and training a model on public MERFISH data to jointly infer molecular locations, cross-round correspondences, and barcodes, alongside cell segmentation and molecule-to-cell assignments, using shared fluorescence and morphology features.

Presentation

Bio-Babel. Oral presentation, Annual Symposium of the PKU Undergraduate Honors Program in Biology, Oct. 2026.

Honors and Awards

Academic Excellence Award, Peking University	2026
Peking University Scholarship	2025
Peking University Freshmen Scholarship	2023
Gold Medal, China National Biology Olympiad Final	2022

Technical Skills

Experimental: In situ spatial omics (STATES), molecular cloning, CRISPR-Cas9, PCR/qPCR, Western blotting.

Computational: Python, R; single-cell and spatial omics analysis, biological image analysis, scientific agent development, workflow automation.